One-page checklist

Chapter 4

- Multiply permutations right-to-left.
- Disjoint cycle decomposition is unique up to order.
- $\text{ord}(\pi) = \text{lcm}(\text{cycle lengths})$.
- A k-cycle has sign $(-1)^{k-1}$.
- To test a group: closure, associativity, identity, inverses.

Chapter 5

- Subgroup test: nonempty and closed under ab^{-1}.
- Cosets partition G; all cosets have size $|H|$.
- Lagrange: $|H| \mid |G|$, so element orders divide $|G|$.
- Isomorphisms preserve identity, inverses, orders, commutativity.
- A code detects $d-1$ errors and corrects $\lfloor (d-1)/2 \rfloor$ errors.

Round 1A: cycle decomposition and inverse

Let

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 \\ 4 & 9 & 10 & 7 & 5 & 2 & 6 & 13 & 1 & 3 & 11 & 12 & 14 & 8 \end{pmatrix}.$$

1. Write π as a product of disjoint cycles.
2. Find π^{-1}.
3. Find $\mathrm{ord}(\pi)$ and $\mathrm{sgn}(\pi)$.

Answer check: Round 1A

$$\pi = (1\ 4\ 7\ 6\ 2\ 9)(3\ 10)(8\ 13\ 14).$$

Hence

$$\pi^{-1} = (1\ 9\ 2\ 6\ 7\ 4)(3\ 10)(8\ 14\ 13).$$

The order is

$$\mathrm{ord}(\pi) = \mathrm{lcm}(6, 2, 3) = 6.$$

The sign is

$$(-1)^{6-1}(-1)^{2-1}(-1)^{3-1} = (-1)^8 = +1.$$

Round 1B: products of cycles

Compute each product, writing the answer as disjoint cycles. Remember: the rightmost cycle acts first.

(i) $(1\ 2\ 3)(2\ 3\ 4)$,

(ii) $(1\ 2\ 3)(3\ 2\ 4)$,

(iii) $(1\ 2\ 3)(3\ 4\ 5)$.

Answer check: Round 1B

$$\begin{aligned}(1\ 2\ 3)(2\ 3\ 4) &= (1\ 2)(3\ 4),\\ (1\ 2\ 3)(3\ 2\ 4) &= (1\ 2\ 4),\\ (1\ 2\ 3)(3\ 4\ 5) &= (1\ 2\ 3\ 4\ 5).\end{aligned}$$

Fast check

Track 1, then the image of 1, and continue until the cycle closes.

Round 1C: order and sign sprint

For each permutation, find its order and sign.

(a) $(1\ 2\ 3\ 4\ 5)(8\ 7\ 6)(10\ 11)$,

(b) $(1\ 2\ 3\ 4)(3\ 4\ 5\ 6)$.

Answer check: Round 1C

	ord	sgn
(a)	$\operatorname{lcm}(5,3,2) = 30$	-1
(b)	$\operatorname{lcm}(3,3) = 3$	$+1$

Round 2A: which are groups?

Decide whether each set is a group under the stated operation. If not, name the failed axiom.

1. $\mathbb{Q}$ under multiplication.
2. $\mathbb{C}^{\times}$ under multiplication.
3. The nonzero integers under multiplication.
4. All functions $\{1, 2, 3\} \to \{1, 2, 3\}$ under composition.
5. $\{a + b\sqrt{2} : a, b \in \mathbb{Z}\}$ under addition.
6. $\mathbb{R}$ under $a * b = a + b + 2$.

Answer check: Round 2A

Case	Group?	Reason
$\mathbb{Q}$ under $\times$	No	0 has no inverse.
$\mathbb{C}^\times$ under $\times$	Yes	usual multiplicative group.
Nonzero integers under $\times$	No	2^{-1} is not an integer.
All functions under composition	No	non-bijections have no inverse.
$\mathbb{Z}[\sqrt{2}]$ under $+$	Yes	identity 0, inverse $-a - b\sqrt{2}$.
$\mathbb{R}$ with $*$	Yes	identity -2, inverse of a is $-a - 4$.

Round 2B: group equations and commutativity

Let G be a group and $a, b \in G$.

1. Prove $(ab)^{-1} = b^{-1}a^{-1}$.
2. Give an example where $(ab)^{-1} \neq a^{-1}b^{-1}$.
3. Suppose $x^2 = e$ for every $x \in G$. Prove that G is Abelian.

Answer check: Round 2B

1. $(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = e$, and similarly on the other side.
2. In $S(3)$, take $a = (1\ 2)$ and $b = (1\ 3)$. Then $ab \neq ba$, so $(ab)^{-1} = b^{-1}a^{-1} = ba \neq ab = a^{-1}b^{-1}$.
3. Since every element is self-inverse, $(ab)^2 = e$. Thus

$$abab = e \quad \Rightarrow \quad ab = (ab)^{-1} = b^{-1}a^{-1} = ba.$$

Hence G is Abelian.

Round 2C: subgroup test

Use the subgroup test to decide whether each is a subgroup.

1. The rotations inside $D(4)$, the symmetries of a square.
2. $\mathbb{R} \setminus \{0\}$ as a subset of $(\mathbb{R}, +)$.
3. $\{\mathrm{id}, (1\ 2), (1\ 3), (2\ 3)\}$ inside $S(3)$.
4. Matrices of the form
$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}, \quad a, b, c \in \mathbb{R},$$
inside $GL(3, \mathbb{R})$.

Answer check: Round 2C

1. Yes. Rotations are closed under composition and inverse.
2. No. Not closed under addition: $1 + (-1) = 0 \notin \mathbb{R} \setminus \{0\}$.
3. No. For example $(1\ 2)(1\ 3) = (1\ 3\ 2)$, not in the set.
4. Yes. The product is again unitriangular, and

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -a & ac - b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix}.$$

Mini-sprint: ring, vector-space checks

Answer quickly: yes/no and one reason.

1. Is $P(X)$ a ring under union as addition and intersection as multiplication?
2. In a ring with no zero-divisors, prove $ac = bc$ and $c \neq 0$ imply $a = b$.

Answer check: mini-sprint

1. No: union has no additive inverses except for $\varnothing$.
2. $ac = bc$ gives $(a - b)c = 0$. Since $c \neq 0$ and there are no zero-divisors, $a - b = 0$, so $a = b$.

Round 3A: cyclic subgroups

Let $G = \langle x \rangle$ be cyclic of order 12.

1. Compute $\langle x^k \rangle$ for $k = 0, 1, 2, 3, 4, 5, 6$. Give its size.
2. State the general rule for $\langle x^k \rangle$ when $\mathrm{ord}(x) = n$.
3. Prove every subgroup of a cyclic group is cyclic.

Answer check: Round 3A

$$|\langle x^k \rangle| = \frac{12}{\gcd(12, k)}.$$

k	0	1	2	3	4	5	6
$\gcd(12, k)$	12	1	2	3	4	1	6
$\lvert\langle x^k \rangle\rvert$	1	12	6	4	3	12	2

General rule: if $\mathrm{ord}(x) = n$, then $\langle x^k \rangle = \langle x^d \rangle$ where $d = \gcd(n, k)$, and its order is n/d.

Proof check: every subgroup of a cyclic group is cyclic

Let $H \leq \langle x \rangle$.

- If $H = \{e\}$, then $H = \langle e \rangle$.
- Otherwise choose the smallest positive m such that $x^m \in H$.
- For any $x^r \in H$, write $r = qm + s$ with $0 \leq s < m$.
- Then
$$x^s = x^r (x^m)^{-q} \in H.$$
  Minimality of m forces $s = 0$.
- Hence every element of H is a power of x^m, so $H = \langle x^m \rangle$.

Round 3B: cosets in G_{14}

Let

$$G_{14} = \{[1], [3], [5], [9], [11], [13]\}$$

be the group of invertible congruence classes modulo 14, and let

$$H = \{[1], [13]\}.$$

1. Write down the distinct left cosets of H.
2. Verify that the cosets partition G_{14}.
3. What does Lagrange's theorem say here?

Answer check: Round 3B

Since G_{14} is Abelian, left and right cosets are the same.

$$H = \{[1], [13]\}, \qquad [3]H = \{[3], [11]\}, \qquad [5]H = \{[5], [9]\}.$$

These three disjoint sets cover G_{14}.

$$|G_{14}| = 6, \quad |H| = 2, \qquad [G_{14} : H] = 3.$$

Lagrange: $2 \mid 6$, and there are exactly $6/2 = 3$ cosets.

Round 3C: element orders in G_{20}

Let
$$G_{20} = \{[1], [3], [7], [9], [11], [13], [17], [19]\}.$$

1. According to Lagrange, what orders are possible?
2. Which of these actually occur?
3. Is G_{20} cyclic?

Answer check: Round 3C

Since $|G_{20}| = 8$, the possible element orders are

$$1, 2, 4, 8.$$

Actual orders:

a	1	3	7	9	11	13	17	19
$\mathrm{ord}([a])$	1	4	4	2	2	4	4	2

No element has order 8, so G_{20} is not cyclic. In fact $G_{20} \cong C_4 \times C_2$.

Round 3D: proof challenge

Let G be a group and define conjugacy by

$$a \sim b \iff b = g^{-1}ag \text{ for some } g \in G.$$

Prove that $\sim$ is an equivalence relation.

Hint

Check reflexive, symmetric, and transitive separately. For symmetry, solve $b = g^{-1}ag$ for a.

Answer check: Round 3D

- Reflexive: $a = e^{-1}ae$, so $a \sim a$.
- Symmetric: if $b = g^{-1}ag$, then $a = gbg^{-1} = (g^{-1})^{-1}b(g^{-1})$, so $b \sim a$.
- Transitive: if $b = g^{-1}ag$ and $c = h^{-1}bh$, then

$$c = h^{-1}g^{-1}agh = (gh)^{-1}a(gh),$$

so $a \sim c$.

Round 4A: isomorphism invariants

For each pair, decide whether the groups can be isomorphic. Give a quick invariant.

1. C_4 and $C_2 \times C_2$.
2. G_8 and $C_2 \times C_2$.
3. $S(3)$ and C_6.
4. $G \times H$ Abelian iff both G and H are Abelian. Prove it.

Answer check: Round 4A

1. Not isomorphic: C_4 has an element of order 4, but $C_2 \times C_2$ does not.
2. $G_8 = \{[1], [3], [5], [7]\}$ and every nonidentity element has order 2, so $G_8 \cong C_2 \times C_2$.
3. Not isomorphic: $S(3)$ is non-Abelian, while C_6 is Abelian.
4. $(g_1, h_1)(g_2, h_2) = (g_1 g_2, h_1 h_2)$. This commutes for all pairs iff $g_1 g_2 = g_2 g_1$ in G and $h_1 h_2 = h_2 h_1$ in H.

Round 4B: direct products

Let $|G| = 6$ and $|H| = 14$.

1. What is $|G \times H|$? What orders are allowed by Lagrange?
2. Which orders can occur for suitable choices of G, H?

Hint

For $(g, h) \in G \times H$,

$$\mathrm{ord}(g, h) = \mathrm{lcm}(\mathrm{ord}(g), \mathrm{ord}(h)).$$

Answer check: Round 4C

$$|G \times H| = 6 \cdot 14 = 84.$$

The possible orders of g divide 6, and the possible orders of h divide 14. Thus

$$\mathrm{ord}(g, h) \in \{\mathrm{lcm}(d_1, d_2) : d_1 \mid 6,\ d_2 \mid 14\}.$$

So the orders allowed in principle are

$$\boxed{1, 2, 3, 6, 7, 14, 21, 42}.$$

All of these occur for suitable choices, for instance $G = C_6$ and $H = C_{14}$. For a fixed noncyclic choice, only a subset may occur.

Round 5A: ISBN check digit

For a 9-digit string $a_1 a_2 \cdots a_9$, the ISBN check digit is chosen so that

$$10a_1 + 9a_2 + \cdots + 2a_9 + \text{check} \equiv 0 \pmod{11}.$$

1. Find the check digit for 0 521 35938.
2. Explain why a single digit error is detected.
3. Explain why swapping two adjacent unequal digits is detected.

Answer check: Round 5A

For 0 521 35938,

$$10a_1 + \cdots + 2a_9 = 0 + 45 + 16 + 7 + 18 + 25 + 36 + 9 + 16 \equiv 7 \pmod{11}.$$

So the check digit is $11 - 7 = 4$.

- A single error of size $k \neq 0$ in position i changes the weighted sum by $k(11 - i)$, nonzero mod 11.
- Swapping adjacent unequal digits u, v changes the sum by $u(11 - i) + v(10 - i) - v(11 - i) - u(10 - i) = u - v$, nonzero mod 11.

Round 5B: minimum distance

Consider the binary linear code generated by

$$G = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

1. List the 8 codewords.
2. Find the minimum distance d.
3. How many errors does the code detect? How many does it correct?

Answer check: Round 5B

The codewords are

$$\begin{array}{cccc} 000000 & 001011 & 010110 & 011101 \\ 100101 & 101110 & 110011 & 111000. \end{array}$$

The nonzero weights are $3, 3, 4, 3, 4, 4, 3$, so

$$d = 3.$$

Therefore the code detects $d - 1 = 2$ errors and corrects

$$\left\lfloor \frac{d-1}{2} \right\rfloor = 1$$

error.

Round 5C: generator matrix challenge

For each generator matrix, find d, the number of detectable errors, and the number of correctable errors.

$$G_1 = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}, \qquad G_2 = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

Answer check: Round 5C

Compute all nonzero codeword weights.

	d	detects	corrects
G_1	2	1	0
G_2	2	1	0

Both codes detect a single error but cannot guarantee correction of a single error.